Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/22/2005
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Project Title:	Countermeasures for Space Radiation Biological Effects		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Radiation Effects Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Philadelphia	**State:**	PA
Zip Code:	19104-6072	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	10/01/2001	**End Date:**	05/31/2004
No. of Post Docs:	3	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-RE00204		
Performance Goal No.:			
Performance Goal Text:			
	It is our hypothesis that control of radiation induced oxidative stress will reduce the risk of cancer development. In this grant work, there are two major specific aims: to determine whether certain dietary supplements can reduce space radiation induced oxidative stress in cultured human cells, and to determine whether the dietary supplements shown to reduce oxidative stress in cultured cells have effects on space radiation induced oxidative stress in animal model systems. The supplemental agents evaluated alone and in combinations include: vitamins C, E, folic acid, glutathione, N-acetyl cysteine (NAC), selenomethionine (SeM), alpha-lipoic acid, niacin, thiamin, Co-enzyme Q10 and the soybean-derived Bowman-Birk inhibitor. The levels of oxidative stress in cultured cells are determined by a dichlorofluorescein fluorescence assay, which has been adapted for evaluation of the effects of space radiations as part of this project. In animals, the total antioxidant status (TAS) in serum or plasma samples is used as the indicator of oxidative stress. The agents shown to reduce space radiation induced oxidative stress in vitro (in MCF10 cells, a human breast epithelial cell		

Task Description:	line) include: vitamin C, vitamin E succinate, NAC, SeM, alpha-lipoic acid (in the reduced form) and Co-enzyme Q10. The effects of treatment with this combination of dietary supplement agents on bio-reduction capacity in animals were evaluated in rats irradiated with iron ions, protons or gamma-rays at a dose of 200 cGy. Sham-irradiated animals were used as controls. The results demonstrated that the TAS decreased in the irradiated animals fed with the control diet. The control animal diet (AIN-93G) contained many of the nutritional supplements at levels comparable on a weight basis to the human RDA levels. Supplementation of the diet with the nutritional supplements prevented the drop in the serum or plasma TAS after the radiation exposure. These results suggest that diet supplementation was effective in preventing the reduction in TAS in the animals irradiated with all three types of radiation. These results indicated that both serum and plasma samples are suitable for a determination of TAS in the irradiated animals. The in vivo and in vitro results described above address the two major specific aims of this research project, and they indicate that the dietary supplement agents evaluated can reduce the levels of radiation induced oxidative stress both in vitro and in vivo. The final year of this project represents a no-cost extension of the grant which ended on May 31, 2004. During the no-cost extension period, projects are being performed to optimize an in vivo assay for the measurement of radiation induced oxidative stress in animal tissues at highest risk for the development of radiation induced cancer (breast, thyroid, etc.). While the assay that has been used for the measurement of oxidative stress in vivo works well for plasma or serum samples, it has not yet been optimized for use in organs like the thyroid. By the end of this no-cost extension of our work, it is hoped that we will have an assay method that has been validated for the measurement of radiation induced oxidative stress in many different organ systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Radiation induced oxidative stress is almost instantaneous with the radiation exposure and thus precedes most, if not all, of the other radiation induced short-term and long-term biological effects. Thus, countermeasures for radiation induced oxidative stress are likely to be very useful for preventing radiation induced tisusue damage, and presumably cancer developpment. The countermeasures for radiation induced oxidative stress developed as part of this research program have significance for all individuals exposed to ionizing radiation on earth as well as in space. The countermeasures shown to protect against radiation induced biological effects from space radiation will be equally useful for protection against radiation induced biological effects on earth.
Task Progress:	Experiments in the following areas were performed during the last funded year. 1) There has been emphasis on applying the results of earlier experiments performed in cultured cells to experiments in whole animals to determine whether the selected dietary supplement agents could have effects on space radiation induced oxidative stress in Sprague-Dawley rats. The effects of treatment with a combination of ascorbic acid, NAC, coenzyme Q10, SeM and vitamin E succinate on bio-reduction capacity in animals were evaluated in rats irradiated with iron ions, protons or gamma-rays at a dose of 200 cGy. Sham-irradiated animals were used as controls. The results demonstated that the TAS decreased in the irradiated animals fed with the control diet. The control animal diet (AIN-93G) contained many of the nutritional supplements at levels comparable on a weight basis to the human RDA levels. Supplementation of the diet with the nutritional supplements prevented the drop in the serum or plasma TAS after the radiation exposure. These results suggest that diet supplementation was effective in preventing the reduction in TAS in the animals irradiated with all three types of radiation. These results indicated that both serum and plasma samples are suitable for a determination of TAS in the irradiated animals. 2) The effects of diet supplementation on radiation induced oxidative stress in rats were also evaluated by measuring the plasma protein carbonyl content, an indicator of protein oxidative damage. The results demonstrated a significant increase in the plasma protein carbonyl content in animals irradiated with 200 cGy and fed with the control diet. In contrast, no increase in the plasma protein carbonyl content was observed in the irradiated rats fed with the supplemented diet. The results suggested that there was increased oxidative stress in the irradiated animals and that the radiation induced oxidative stress can be prevented by treatment with the nutritional supplement agent combination. 3) The effects of the dietary supplement agents were determined on radiation induced transformation in HTori-3 cells, a line of human thyroid epithelial cells. Treatment with the dietary supplement agents prevented iron ion and proton radiation induced malignant transformation of HTori-3 cells. 4) The effects of the nutritional supplements on radiation induced cytotoxicity of MCF10 cells were determined by a clonogenic survival assay. It was observed that the nutritional supplements could reduce cytotoxicity resulting from exposure to radiation from 1 or 5-GeV/nucleon iron ions. 5) While none of the agents evaluated by themselves can completely prevent/suppress the biological phenomenon studied in this project, when tested together as a combination, they worked together to have a highly significant suppressive effect on the adverse biological effects of space radiation that were studied, which included malignant transformation and cytotoxicity.
Bibliography Type:	Description: (Last Updated: 08/24/2020)
Articles in Peer-reviewed Journals	Kennedy AR, Ware JH, Guan J, Donahue JJ, Biaglow JE, Zhou Z, Stewart J, Vazquez M, Wan XS. "Selenomethionine protects against adverse biological effects induced by space radiation." Free Radic Biol Med. 2004 Jan 15;36(2):259-66. PMID: 14744637 , Jan-2004
Articles in Peer-reviewed Journals	Wan XS, Zhou Z, Kennedy AR. "Adaptation of the dichlorofluorescein assay for detection of radiation-induced oxidative stress in cultured cells." Radiat Res. 2003 Dec;160(6):622-30. PMID: 14640785 , Jan-2004
Articles in Peer-reviewed Journals	Guan J, Wan XS, Zhou Z, Ware J, Donahue JJ, Biaglow JE, Kennedy AR. "Effects of dietary supplements on space radiation-induced oxidative stress in Sprague-Dawley rats." Radiat Res. 2004 Nov;162(5):572-9. PMID: 15624312 , Nov-2004
Awards	Kennedy AR. "Discipline Co-Chair (with Dr. Aloke Chatterjee), Radiation Section, Bioastronautics Investigators' Workshop, Galveston, Texas, January 17-19, 2001. " Jan-2004
Awards	Kennedy AR. "Discipline Co-Chair (with Dr. Amy Kronenberg), Radiation Section, Bioastronautics Investigators' Workshop, Galveston, Texas, January 14, 2003. " Jan-2004
Awards	Kennedy A. "Invited Participant, - NASA/NSBRI Radiation Steering Group Meeting, Brookhaven National Laboratory, Upton, New York, May 5, 2003. " Jan-2004

Awards	Kennedy A. "Invited Participant, NASA/NSBRI Working Group on Standardized Flight Data Collection, July 31-August 2, 2001, Del Lago Conference Center, Conroe, Texas. " Jan-2004
Presentation	Kennedy, A. R. "Cancer Prevention Through Reduction of Oxidative Stress During Space Travel " Oct-2002
Presentation	Kennedy, A. R. "Countermeasures for Space Radiation Biological Effects " Jan-2003
Presentation	Kennedy, A. R. and A. Chatterjee "Summary of Radiation Studies " Jan-2001